Module 09: Polymorphism and Types

Objective: Dealing with polymorphism and types

System requirements: .Net 5.0

Tools: Visual Studio 2019

Duration: ~30min

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1 Extension of the vehicle class

- Extend the vehicle class from the previous exercises with an abstract method Honk(). Implement it in the inheriting classes (e.g. by a console output or similar).
- Extend the vehicle class with the method
  public static Vehicle GenerateVehicle(string nameSuffix = "")
  which should randomly generate a standard car, ship or aircraft. Use the .Next() method of an object of the System.Random class. By means of the passing parameter the respective default name shall be extended.
- Let the vehicle class override the ToString() method, so that the type and the name of the vehicle are now output.

2 Vehicle-Array

- Create a Vehicle[] array of length 10 in the Main() method.
- Fill this with cars, planes and ships, creating these objects using the new random method of the vehicle class.

3 Type counting

- Output the ToString() methods of the generated vehicles.
- Now count how many objects of each type are in the array and output the counts via the console.
- Let the vehicle at the 3rd array position honk.

```
We have produced 5 cars, 2 planes, and 3 ships.
Car: Beeeeep
```